

- Description

This module manages a temperature control system for Air Conditioner, ensuring that:

1. The temperature **must be within the valid operating range** (e.g., 16°C to 23°C).
2. If the temperature exceeds the upper limit, **a cooling mechanism is activated**.
3. If the temperature falls below the lower limit, **a heating mechanism is activated**.
4. If the temperature is within the acceptable range, **no action is taken**.

This ensures that the system maintains a stable temperature within the specified limits.

- Specification

module TemperatureControlSystem

var temperature: real

process setTemperature(newTemp: real)

functional scenario form:

newTemp < 16 && Warning: Temperature too low! Activating heating system...

||

newTemp > 23 && Warning: Temperature too high! Activating cooling system...

||

newTemp >= 16 && newTemp <= 23 && Temperature is stable. No action required.

end process

end module

- Java Code

```
package code;
public class c05_TemperatureController {
    private double temperature;
    private final double MIN_TEMP = 16.0;
    private final double MAX_TEMP = 23.0;
    public c05_TemperatureController(double initialTemp) {
        this.temperature = initialTemp;
    }
    public void setTemperature(double newTemp) {
        this.temperature = newTemp;
        if (newTemp < MIN_TEMP) {
            System.out.println("Warning: Temperature too low! Activating heating system...");
        } else if (newTemp > MAX_TEMP) {
            System.out.println("Warning: Temperature too high! Activating cooling system...");
        } else {
            System.out.println("Temperature is stable. No action required.");
        }
    }
}
```

```

    }
    public double getTemperature() {
        return temperature;
    }
    public static void main(String[] args) {
        // testcase-VT:
        //      c05_TemperatureController controller1 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller1.setTemperature(18.3);    //newTemp >= 16 && newTemp
        //      <= 23
        //      c05_TemperatureController controller2 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller2.setTemperature(37.433);    //newTemp > 23
        //      c05_TemperatureController controller3 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller3.setTemperature(0);    //newTemp < 16

        // testcase-FT:
        //      c05_TemperatureController controller1 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller1.setTemperature(-1.7976931348623157E308);
        //      c05_TemperatureController controller2 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller2.setTemperature(1.1209145429141497E308);
        //      c05_TemperatureController controller3 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller3.setTemperature(0);

        // testcase-Z3:
        //      c05_TemperatureController controller1 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller1.setTemperature(16);    //newTemp >= 16 && newTemp <
        //      = 23
        //      c05_TemperatureController controller2 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller2.setTemperature(15);    //newTemp > 23
        //      c05_TemperatureController controller3 = new c05_TemperatureContr
        //      oller(20.0);
        //      controller3.setTemperature(24);    //newTemp < 16

        // testcase-FT:
        c05_TemperatureController controller1 = new c05_TemperatureContro
        ller(20.0);
        controller1.setTemperature(8);

```

```

        c05_TemperatureController controller2 = new c05_TemperatureContro
ller(20.0);
        controller2.setTemperature(308);
        c05_TemperatureController controller3 = new c05_TemperatureContro
ller(20.0);
        controller3.setTemperature(20);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	812	816
✓ src	0.5 %	4	812	816
✓ code	0.5 %	4	812	816
✓ c05_TemperatureController.java	100.0 %	4	0	4
✓ c05_TemperatureController	100.0 %	4	0	4
main(String[])		0	0	0
c05_TemperatureController(dou		0	0	0
setTemperature(double)	100.0 %	4	0	4

3. Vibration testing instruction coverage: 95.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.4 %	60	15,450	15,510
✓ src	0.4 %	60	15,450	15,510
✓ code	0.4 %	60	15,450	15,510
✓ c05_TemperatureController.java	95.2 %	60	3	63
✓ c05_TemperatureController	95.2 %	60	3	63
main(String[])	100.0 %	25	0	25
c05_TemperatureController(dou	100.0 %	12	0	12
setTemperature(double)	100.0 %	23	0	23

4. Fuzz testing branch coverage: 75%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.4 %	3	813	816
✓ src	0.4 %	3	813	816
✓ code	0.4 %	3	813	816
✓ c05_TemperatureController.java	75.0 %	3	1	4
✓ c05_TemperatureController	75.0 %	3	1	4
setTemperature(double)	75.0 %	3	1	4
main(String[])		0	0	0
c05_TemperatureController(dou		0	0	0

5. Fuzz testing instruction coverage: 90.5%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.4 %	57	15,453	15,510
✓ src	0.4 %	57	15,453	15,510
✓ code	0.4 %	57	15,453	15,510
✓ c05_TemperatureController.java	90.5 %	57	6	63
✓ c05_TemperatureController	90.5 %	57	6	63
setTemperature(double)	87.0 %	20	3	23
main(String[])	100.0 %	25	0	25
c05_TemperatureController(dou	100.0 %	12	0	12

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c05_TemperatureController.java	100.0 %	4	0	4
▼ c05_TemperatureController	100.0 %	4	0	4
setTemperature(double)	100.0 %	4	0	4
main(String[])		0	0	0
c05_TemperatureController(double)		0	0	0

7. Z3 instruction coverage: 95.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	60	22,026	22,086
▼ src	0.3 %	60	22,026	22,086
▼ code	0.3 %	60	22,026	22,086
▼ c05_TemperatureController.java	95.2 %	60	3	63
▼ c05_TemperatureController	95.2 %	60	3	63
main(String[])	100.0 %	25	0	25
c05_TemperatureController(double)	100.0 %	12	0	12
setTemperature(double)	100.0 %	23	0	23

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.5 %	4	836	840
▼ src	0.5 %	4	836	840
▼ code	0.5 %	4	836	840
▼ c05_TemperatureController.java	100.0 %	4	0	4
▼ c05_TemperatureController	100.0 %	4	0	4
setTemperature(double)	100.0 %	4	0	4
main(String[])		0	0	0
c05_TemperatureController(double)		0	0	0

9. Unimproved V-method instruction coverage: 95.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	60	22,026	22,086
▼ src	0.3 %	60	22,026	22,086
▼ code	0.3 %	60	22,026	22,086
▼ c05_TemperatureController.java	95.2 %	60	3	63
▼ c05_TemperatureController	95.2 %	60	3	63
main(String[])	100.0 %	25	0	25
c05_TemperatureController(double)	100.0 %	12	0	12
setTemperature(double)	100.0 %	23	0	23